In the name of God

School of Electrical and Computer Engineering		Spring 1404
Reinforcement Learning		Computer Assignment # 3

Post Delivery Robot Simulation using Monte Carlo

In this assignment, you will simulate a post-delivery robot that must **pick up a package at one location** within a gridworld environment. **The episode ends when the package is successfully picked up.**

The environment is a 5×5 grid.
The robot can move: North, South, East, West, and also Pick Up packages.

Rewards:

- Successful Pick Up: **+20**
- Illegal move*: **-10**
- Every move: **-1**

* Illegal moves:

- Pick-up when no package is at the robot's location
- Move into a wall

At the start of each episode:

- The package is randomly positioned at one **of the four marked cells** (Figure 1)
- The robot starts from a **random location** on the grid

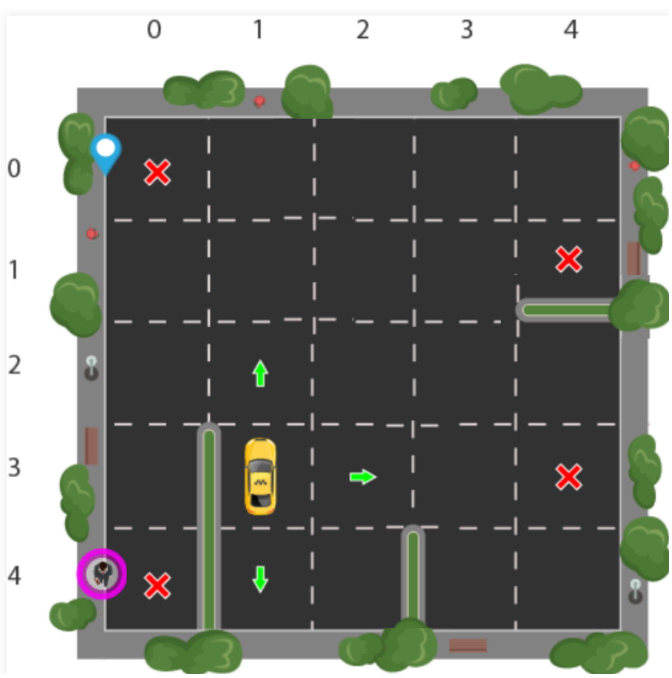

Figure 1: A 5×5 gridworld. The four marked cells indicate possible pickup locations for the package.

Part 1: Environment Setup

Model the environment as a **5×5 gridworld**.
Define the action space and the state space
Initialize the environment at the start of each episode:

- A **package** is placed at a randomly selected **marked cell in Figure 1**

Episode ends when the package is picked up **(or terminated after 25 time steps)**.
Note: The robot is aware of the location of the package at the start of each episode.
Handle rewards and penalties according to the problem description.

Part 2: Solve Using On-Policy Monte Carlo

Implement an **on-policy** first-visit Monte Carlo control method to learn the optimal policy.

Part 3: Solve Using Off-Policy Monte Carlo

Implement an **off-policy** Monte Carlo control method using **importance sampling**.
Use a behavior policy (e.g., more exploratory ε-greedy) and a separate target policy.

Required Outputs (for 100,000 episodes):

1. Plot the **average return per episode overtime** for Part 2 and Part 3.
2. Show the learned **optimal policy** by plotting the best action from each grid position. (You may need to report 4 figures for 4 possible package locations)
3. Plot the estimated **state-value function (V(s))** after training is complete for each method.(You may need to report 4 figures for 4 possible package locations)

Submission Requirements:

- The **source code** for the environment and agents.
- Any plots you generate.
- A **short report** explaining your approach and findings, and instructions on how to run your code.
